Active Directory: Group Basics

Groups in **Active Directory (AD)** are used to simplify the management of permissions and access. Instead of assigning permissions to individual users, you assign permissions to groups and then add users to those groups.

Benefits of Using Groups

- Simplifies permission management.
- Reduces administrative effort.
- Makes access control more consistent.
- Supports role-based access (e.g., HR, Finance, IT).

Types of Groups

1. Security Groups

- Used to assign permissions to resources such as files, folders, printers, and applications.
- Can also be used as email distribution lists if needed.
- Most commonly used group type.

Example:

- Sales_Read
- HR_Modify
- IT_Admins

---

2. Distribution Groups

- Used only for email distribution.
- Cannot be used to assign security permissions.
- Typically used with Microsoft Exchange.

Example:

- All Employees

- Marketing Team

Group Scopes

Domain Local

- Used to assign permissions to resources within the same domain.
- Can contain:
    - Users
    - Global groups
    - Universal groups
    - Other domain local groups (from the same domain)

Example:
A **FileServer_Read** group that grants access to a shared folder.

Global

- Used to group users with similar job roles from the same domain.
- Can contain:
    - Users
    - Other global groups from the same domain
- Can be assigned permissions in any domain within the forest.

Example:
A **Sales** group containing all sales employees.

Universal

- Used in multi-domain forests.
- Can contain users and groups from any domain.
- Can be granted permissions anywhere in the forest.

Example:
A **Corporate Managers** group containing managers from multiple domains.

Group Nesting

A common best practice is to nest groups rather than assigning permissions directly to users.

Example:

User → Global Group → Domain Local Group → Resource

For example:

John Smith

|

▼

Sales (Global Group)

|

▼

Sales_Share_RW (Domain Local Group)

|

▼

Shared Folder

This approach makes permission management easier as users join, leave, or change roles.

Managing Groups

Create a Group

1. Open **Active Directory Users and Computers (ADUC)**.
2. Navigate to the appropriate Organizational Unit (OU).
3. Right-click the OU and select **New → Group**.
4. Enter:

- Group name
- Group scope (Domain Local, Global, or Universal)
- Group type (Security or Distribution)

5. Click **OK**.

Add Members

1. Open the group properties.
2. Select the **Members** tab.
3. Click **Add**.
4. Select users or other groups.
5. Click **OK**.

Remove Members

1. Open the **Members** tab.
2. Select the user or group.
3. Click **Remove**.

PowerShell Examples

Create a Security Group

```
New-ADGroup `
  -Name "Sales" `
  -SamAccountName "Sales" `
  -GroupCategory Security `
  -GroupScope Global `
  -Path "OU=Groups,DC=contoso,DC=com"
```

Add a User to a Group

Add-ADGroupMember -Identity "Sales" -Members "jdoe"

Remove a User from a Group

Remove-ADGroupMember -Identity "Sales" -Members "jdoe"

View Group Members

Get-ADGroupMember -Identity "Sales"

Best Practices

- Use **Security groups** for assigning permissions.
- Assign permissions to **groups**, not directly to individual users.
- Use descriptive, consistent group names (for example, HR_Read, Finance_Modify, IT_Admins).
- Keep group membership up to date by removing users who no longer require access.
- Use **group nesting** to simplify permission management in larger environments.
- Follow the **AGDLP** model where appropriate:
    - **A**ccounts → **G**lobal Groups → **D**omain **L**ocal Groups → **P**ermissions. This helps separate user management from resource permission management and scales well in enterprise environments.